Advanced Programming Practice · 21CSC203P

Unit 1 · Session 1 · SLO 2 — Solved Worksheet

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Unit – 1 : Session – 1 : SLO – 2 | Language Classification

Activity (Solutions)

• 1) Java is BOTH compiled and interpreted (hybrid). The javac compiler translates source code (.java) into platform-independent bytecode (.class); the JVM then interprets this bytecode at run time (with JIT compilation to native machine code for speed). So Java is compiled to bytecode and interpreted/JIT-compiled by the JVM — this gives portability ('write once, run anywhere').

• 2) Simple program: public class Hi { public static void main(String[] a){ System.out.println("Hello"); } } — Conversion: javac Hi.java compiles the source into Hi.class containing bytecode instructions (e.g., getstatic, ldc, invokevirtual). The JVM's class loader loads Hi.class, the bytecode verifier checks it, and the interpreter/JIT executes it as machine code.

• 3) Low-level vs high-level: Low-level languages (machine code, assembly) are close to hardware — one instruction maps to one machine operation, fast but hard to write and not portable. High-level languages (Java, Python, C++) use English-like abstractions, are portable and easier to read/write, and require a compiler/interpreter to translate them into machine code.